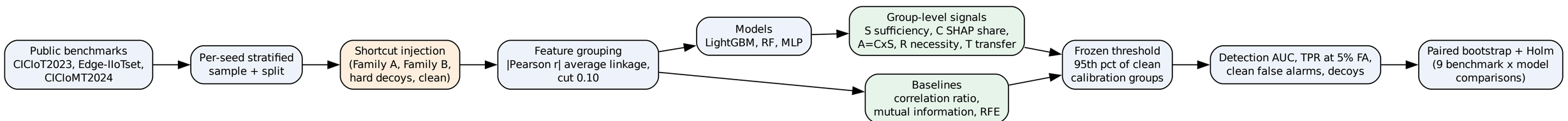
Public benchmarks
CICIoT2023, Edge-IIoTset,
CICIoMT2024
Per-seed stratified
sample + split
Shortcut injection
(Family A, Family B,
hard decoys, clean)
Feature grouping
|Pearson r| average linkage,
cut 0.10
Models
LightGBM, RF, MLP
Group-level signals
S sufficiency, C SHAP share,
A=CxS, R necessity, T transfer
Baselines
correlation ratio,
mutual information, RFE
Frozen threshold
95th pct of clean
calibration groups
Detection AUC, TPR at 5% FA,
clean false alarms, decoys
Paired bootstrap + Holm
(9 benchmark x model
comparisons)